

FCU Development and FADEC Integrated Testing on Cube Orange

1. Objective

The objective of this activity is to develop, integrate, simulate, verify, and validate the Flight Control Unit (FCU) using Cube Orange hardware running ArduPilot. The FCU is integrated with the FADEC system through CAN communication and validated through Software-in-the-Loop (SIL), Hardware-in-the-Loop (HIL), and ground testing.

Hardware Setup

FCU Hardware

- Cube Orange Flight Controller installed.
- CUAV NEO 3 GPS connected to GPS2 Port.
- RM3100 Compass verified through Mission Planner.
- Telemetry Radio connected to TELEM1.
- Servos connected to MAIN OUT ports.
- CAN Transceiver connected between Cube Orange and FADEC.
- USB Joystick connected to Ground Control Station.



Mission Planner Configuration

GPS Configuration

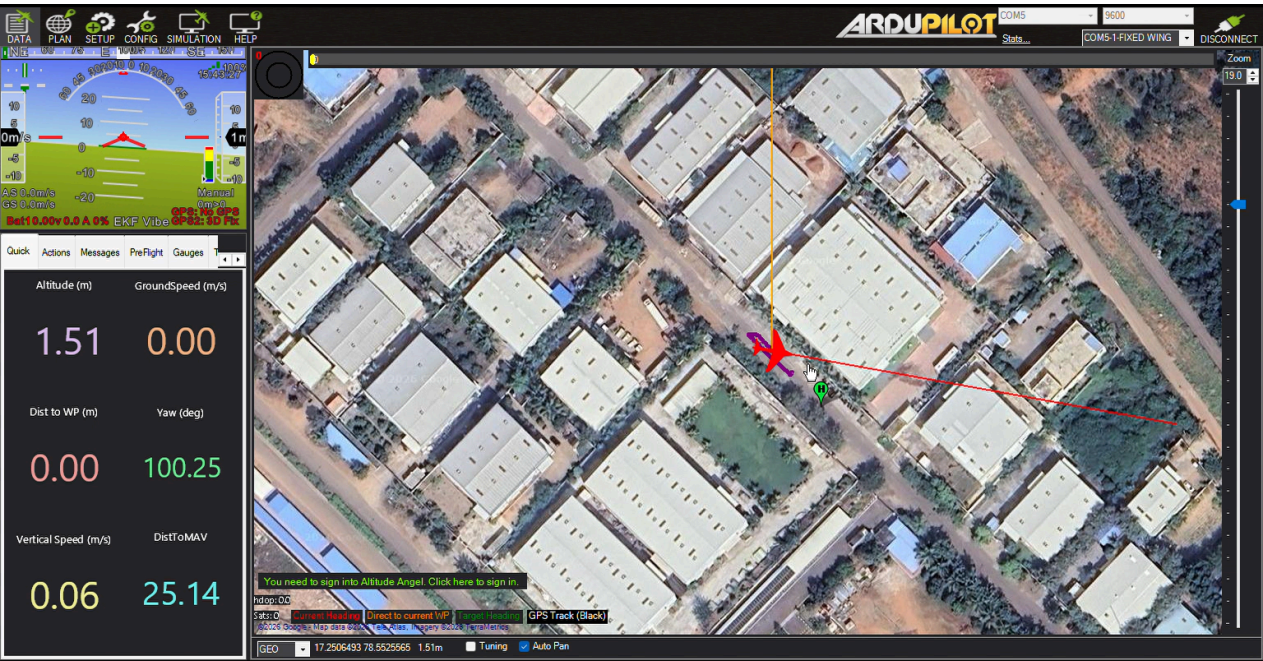
Parameter	Value
GPS_TYPE2	1
SERIAL4_PROTOCOL	5
SERIAL4_BAUD	115

CAN Configuration

Parameter	Value
CAN_P1_DRIVER	1
CAN_D1_PROTOCOL	1
CAN_P1_BITRATE	1000000

Flight Modes

Mode	Function
AUTO	Mission Execution
GUIDED	Command Testing
LOITER	Position Hold
RTL	Recovery
MANUAL	Ground Validation



🚩 Mission Planner 1.3.83 build 1.3.9384.38258 ArduPlane V4.6.3 (3fc701...

Servo Validation

Output	Function
MAIN OUT 1	Aileron
MAIN OUT 2	Elevator
MAIN OUT 3	Throttle
MAIN OUT 4	Rudder
MAIN OUT 8	Parachute Trigger

Servo Parameters

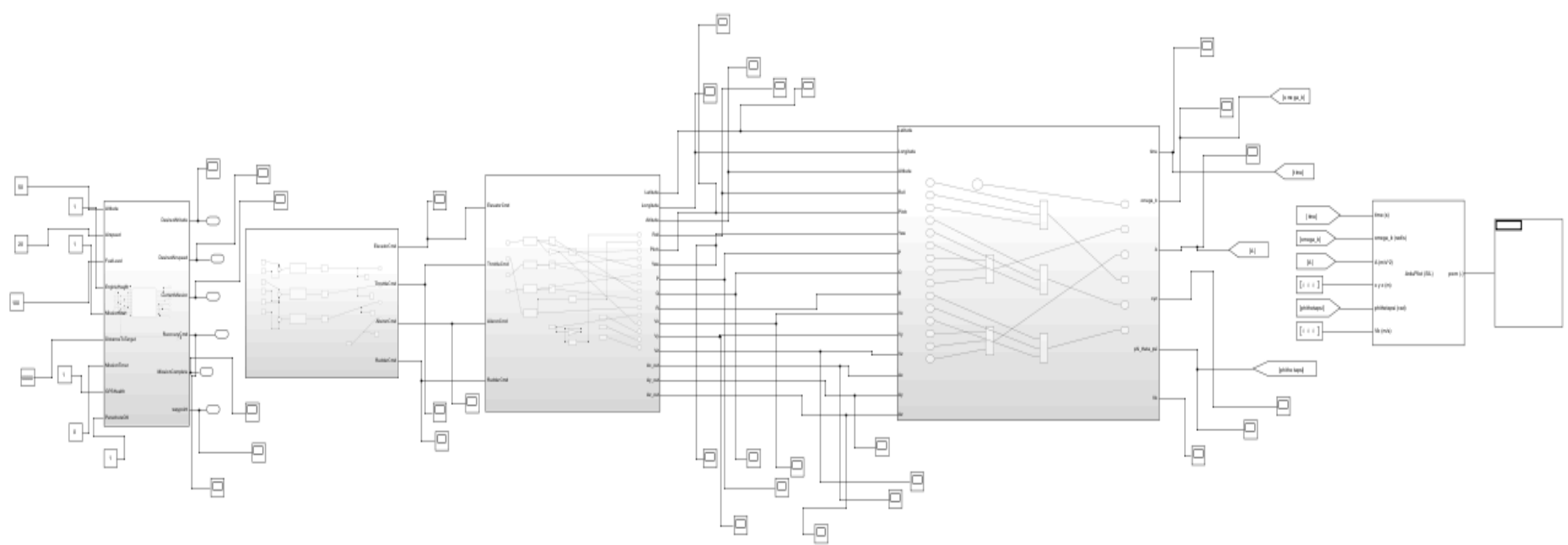
Parameter	Value
SERVO1_MIN	1000
SERVO1_MAX	2000
SERVO1_TRIM	1500
SERVO8_FUNCTION	1

Parameter	Value
SERIAL4_PROTOCOL	GPS
SERIAL4_BAUD	115200
GPS_TYPE	Auto
COMPASS_ENABLE	1
COMPASS_USE	1
COMPASS_CAL_FIT	16
COMPASS_LEARN	3
BARO_ALT_OFFSET	0
EK3_SRC1_POSZ	1
ARMING_REQUIRE	1
ARMING_CHECK	1

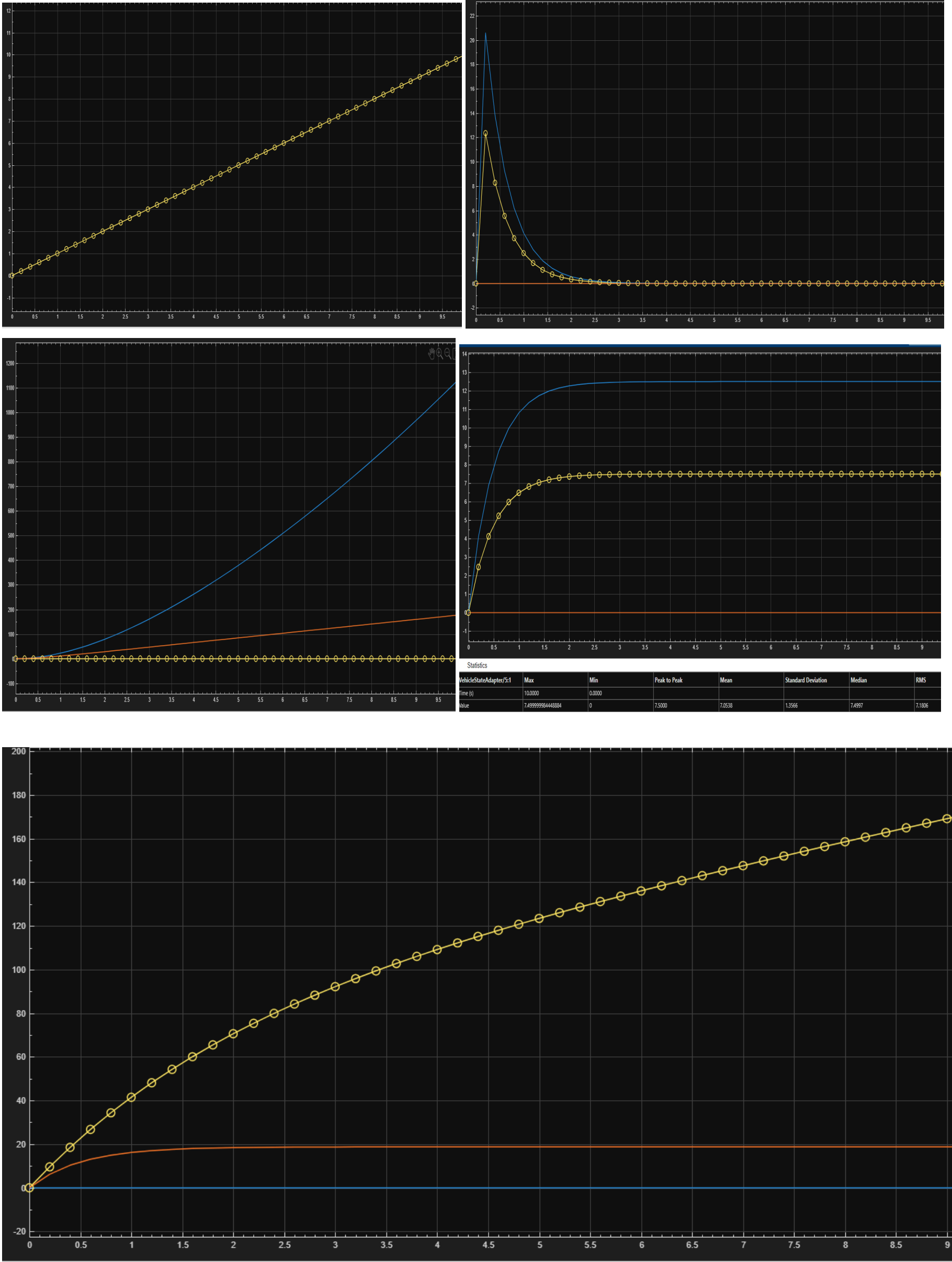
Parameter	Value
GPS_TYPE	Auto
COMPASS_ENABLE	1
COMPASS_USE	1
COMPASS_CAL_FIT	16
COMPASS_LEARN	3
BARO_ALT_OFFSET	0
EK3_SRC1_POSZ	1

For commands to connect with simulink :

<https://fst.aviumtechnologies.com/ardupilot-sil-connector>

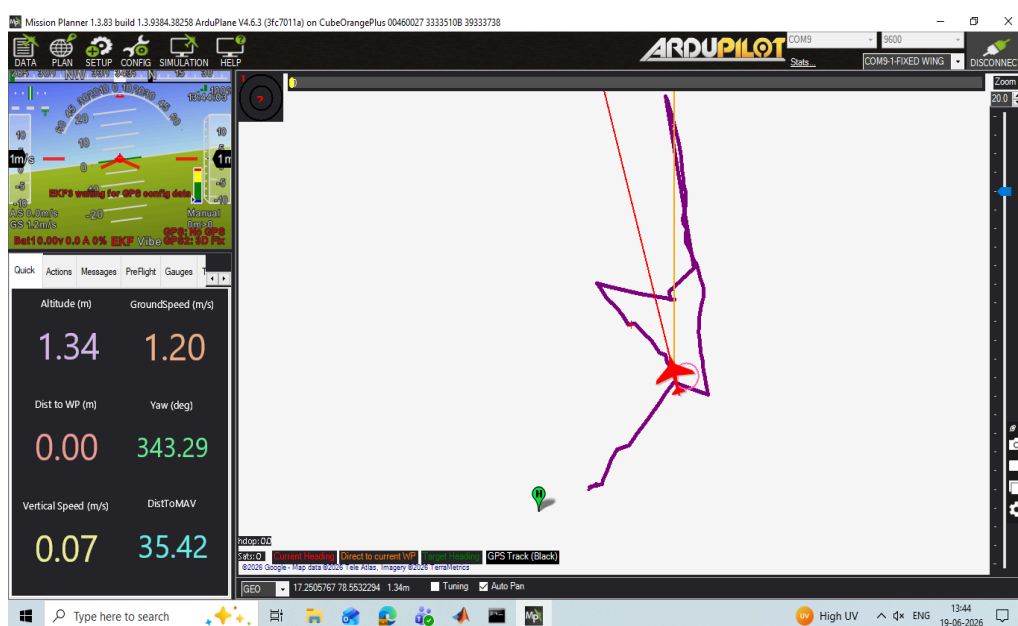
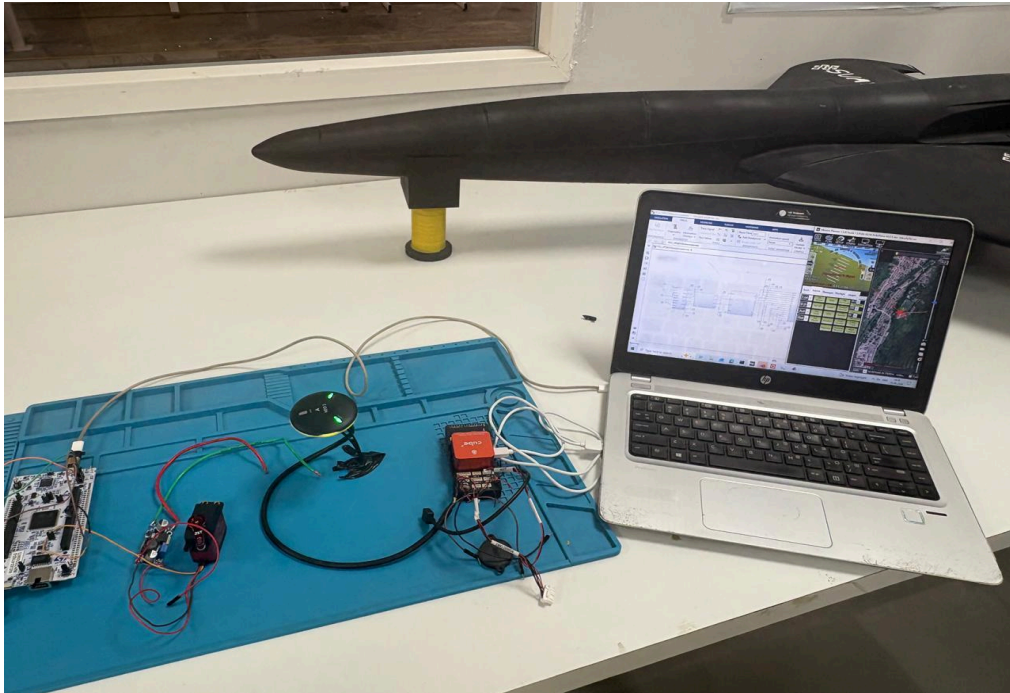


RESULTS:



Demo video:

[missionplannersimulink.mp4](#)



Ardupilot Setup HITL & SITL (Simulink)

Step 1 : Requirements and Downloads

Install MATLAB-supported compiler

<https://mathworks.com/support/requirements/supported-compilers.html>

You can install MinGW compiler for Simulink rather than Visual Studio file

Make sure to follow same folder structure and files

<https://fst.aviumtechnologies.com/ardupilot-sil-connector>

Extract the includes in same folder

Commands :-

1) In Matlab

```
[status,out] = system('where g++');
```

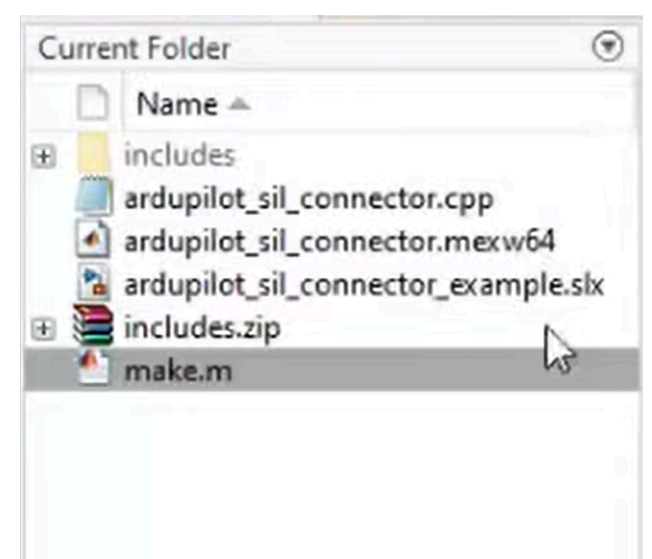
```
disp(out)
```

2) mex -setup C++

```
mex.getCompilerConfigurations('C++')
```

3) mex -I./includes ardupilot_sil_connector.cpp ws2_32.lib

4) mex-setup



5) make

Step 2 : Mission Planner and Settings

1) Download and install MissionPlanner

<https://ardupilot.org/planner/docs/mission-planner-installation.html>.

2) Open MissionPlanner, select the "Simulation" tab and click on the "Plane" icon to download the ArduPlane SIL simulator

3) Go to folder location "C:\Users\<username>\Documents\Mission Planner\sitl" and type cmd to open CMD

4) Execute the following command in the command prompt:

```
ArduPlane.exe --home 42.841448,24.770727,0.5,0 --model json
```

5) Click connect icon in Mission Planner

Connect Type : - TCP

Baud Rate :- 56000

Connect

Same laptop then IP address is 127.0.0.1

Different laptop use there Laptop IP instead of 127.0.0.1

Remote Port 5760 (dont change)

Note : 9002 port is for UDP connection for Simulink to Ardupilot tool and 5760 is for connection port of Ardupilot to Mission Planner

6) Open and Run you simulink model (Simulink Ardupilot S function must give 6 inputs)

Note : Make the Simulink model fixed-step, not variable-step

Steps :

a) Go to Modeling - Model

b) Select Solver

c) Type: Fixed-step

Solver: ode4 (Runge-Kutta)

Fixed-step size: 0.01